ENTRY QUESTIONS TO MODERN PHYSICS EXAM

Physical constants

$m_e c^2 = 511\,\text{keV}, \qquad hc = 1240\,\text{eV nm}$

Questions

1. Draw the setup to measure the photoelectric effect in detail (pay attention to polarities, current directions, etc.), and sketch the $I-V$ curves.
2. You use a Laser of wavelength $\lambda = 500\,\text{nm}$. Compute the threshold voltage in the photoelectric effect, assuming Cs electrodes with a work function $\phi = 1.95\,\text{eV}$.
3. Write down Einstein's and de Broglie's formula, and derive the dispersion relation, $\omega(k) = \hbar k^2/(2m_e)$ for an electron. Prove that the group velocity of the matter wave is just $v_g = p/m_e$.
4. Sketch Bragg scattering (geometry of scattering on a crystal), and explain the Bragg condition.
5. Consider an electron in a one-dimensional box of size L. Write down its normalized ground state wave function, $\psi_1(x)$, and show that it is an eigenstate of the time-independent Schrödinger equation with eigenvalue $E_1 = h^2/(8m_e L^2)$. What is this energy for $L = 1\,\text{nm}$?
6. Derive the time-independent one-dimensional Schrödinger equation from the time-dependent Schrödinger equation.
7. Write down the following operators: $\hat{x}$, $\hat{p}_x$, $\hat{H}$ for a particle of mass M moving in a parabolic potential $V(x) = Dx^2/2$. Write down the one-dimensional time-independent Schrödinger equation.
8. Express the eigenenergies, and sketch graphically the first three eigenstates of a harmonic oscillator characterized by a mass M and spring constant D.
9. Write down the structure of the Hydrogen wave function $\psi_{211}(r,\theta,\phi)$ of the $2p$ level, and sketch its probability distribution. What is the energy of this wave function (in electronvolts), and its angular momentum around the z axis (in units of $\hbar$)?
10. Enumerate the four quantum numbers characterizing an electron state in the Hydrogen atom, and list their possible values. Compute the total degeneracy of the L shell!
11. Consider the $2p$ state of Hydrogen. What is the value of L and of S, and the total degeneracy of this state in the absence of spin-orbit (LS) coupling? What are the possible values of J in the presence of spin-orbit coupling? Which one is lower in energy? Discuss degeneracies!
12. Consider an isolated Al atom ($Z = 13$). Write down/discuss its shell structure. What are the ground state values of S, L, and J? What is the ground state degeneracy in the absence of external field?
13. Derive the velocity addition formula from the Lorentz transformation $(x,t) \to (x',t')$ (given above) connecting the velocities u and u' measured in the rest frame and in the frame moving with a velocity v.
14. Write down the equations of time dilation and length contraction, and explain them.
15. What voltage do you need to accelerate an electron to a velocity $v = (4/5)\,c$?
16. In 1996 the United States had only 75 kg of tritium stockpiled. If none of that was used, how much tritium remained in 2020 ($t_{1/2} = 12\,\text{years}$)? What was the activity of this tritium in 2020?
17. Draw and explain the way a linear accelerator works.
18. Draw the Feynman diagram of the β^- decay. Specify all particles in the diagram, and write down the corresponding reaction. List the various quantum numbers of the particles (charge, spin, lepton number, baryon number).
19. Draw the Feynman diagram of electron-positron annihilation. What is the energy of the outcoming photons if the electron and the positron are at rest?
20. What is the quark content of the following particles: p, e^+, $\overline{n}$, π^+? Explain their charge in terms of the quark charges.